

Bicarbonate Inhibition of Carbonic Anhydrase Mimics Hinders Catalytic Efficiency: Elucidating the Mechanism and Gaining Insight toward Improving Speed and Efficiency

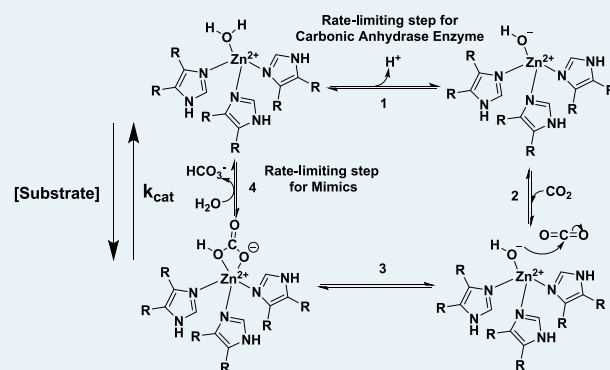
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Supporting Information

ABSTRACT: Carbonic anhydrase (CA) mimics are often studied with a focus on the hydration of CO₂ for atmospheric carbon capture. Consequently, the reverse reaction (dehydration of HCO₃[−]) has received minimal attention, so much so that the rate-limiting step of the dehydration reaction in CA mimics is currently unknown. The rate-limiting step of the hydration reaction is reported to be the bicarbonate-bound intermediate step, and thus is susceptible to product inhibition. It is not, however, clear if this inhibition is a consequence of an increase in the rate of the competing dehydration reaction or resulting from the strong affinity of bicarbonate to the mimic. To address this, insight into the dehydration reaction kinetics is needed. We therefore report the most comprehensive study of a CA mimic to date. The dehydration profile of the fastest small-molecule CA mimic, ZnL1S, was characterized, and consequently evidence for the rate-limiting step for the dehydration reaction was seen to be the bicarbonate-bound intermediate step, much like the hydration reaction. This experimental validation of the rate-limiting step was achieved through a variety of methods including NMR experiments and the effect of inhibitors, substrate concentration, and metal center on activity. With this understanding, an improvement in the favorability of the rate-limiting step was achieved, resulting in decreased bicarbonate inhibition. Thus, an increase in the mimic's k_{cat} for both reactions was observed, resulting in the largest rate constants of any small-molecule CA mimic reported to date (28 093 and 579 M^{−1} s^{−1} for hydration and dehydration, respectively). Enzyme-like $k_{\text{cat}}/k_{\text{m}}$ values were obtained for ZnL1S (5.9 × 10⁵ M^{−1} s^{−1} for CO₂ hydration), and notably there is only a difference of 2.5 orders of magnitude from the enzyme, the closest of any CA mimic reported in the literature. The results from this work can be applied to the development and improvement of future and existing mimics toward attaining increased activities.

KEYWORDS: carbonic anhydrase, biomimetic, HCO₃[−] dehydration, CO₂ hydration, rate-limiting step, stopped-flow



INTRODUCTION

The capture of CO₂ produced during industrial processes, or directly from the atmosphere, is an important process and has received significant attention to reduce the impact of climate change resulting from increased atmospheric CO₂.^{1–3} Frequently, synthetic mimics of the enzyme carbonic anhydrase (CA),^{4–6} which catalyze the reversible reaction between carbon dioxide hydration and bicarbonate dehydration, are investigated as CO₂ capture agents, for storage or other uses.^{3,7–15} Most synthetic CA mimics replicate the active site of the α class of mammalian CA's and thus are composed of a heteromacrocycle scaffold coordinating zinc,^{16–19} and are often designed for use in organic solvents.⁹ There are reports of CA mimics that are catalytically active in aqueous solutions at rates similar to those that work only in organic solvents, albeit with rates around 4–5 orders of magnitude slower than that of the natural enzyme.^{18,20–23}

An example of a CA mimic, ZnL1S (Chart 1), works in aqueous environments and possesses one of the fastest rates of CO₂ hydration to date.²⁴ It coordinates a zinc ion through a scaffold containing three histidine-like imidazole ligands and a water molecule. Additionally, the authors state that the ligand generates a hydrophobic pocket within the structure, facilitating the direction of CO₂ approach to the reactive zinc center, thereby overcoming the activation barrier of CO₂ hydration and thus increasing hydration rates.²⁴ While the CA mimics and the natural CA enzyme have the same catalytic mechanism (Scheme 1),^{4,25} their kinetics differ significantly.

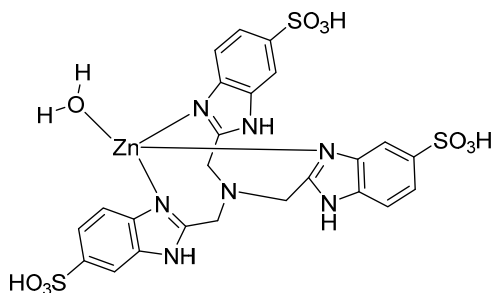
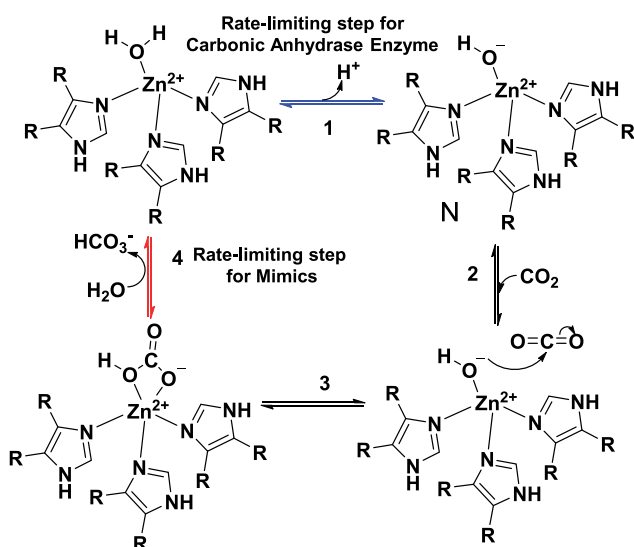
The rate-limiting step of the natural enzyme is the deprotonation of the bound water molecule, which, due to

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Chart 1. ZnL1S, a Water-Soluble Small-Molecule CA Mimic

Scheme 1. Mechanism of Carbonic Anhydrase and Its Mimics^a

^aStep 1 (blue): Deprotonation of a zinc-bound water to generate (His)₃–Zn–OH[–]. Step 2: Nucleophilic attack of CO₂. Step 3: Bicarbonate-bound intermediate formed. Step 4 (red): Displacement of bicarbonate by water to generate the complex initial state. The R groups on the zinc-coordinating imidazole's represent either an attached benzene moiety to form benzimidazole, like in ZnL1S,²⁴ or part of a histidine amino acid, as is the case for the active site of the mammalian α -carbonic anhydrase.⁴

the shuttling of protons, is essentially diffusion limited (step 1 in Scheme 1 highlighted in blue).²⁶ For this reason, it has one of the highest known enzymatic catalytic activities, with a rate of CO₂ hydration of $\sim 10^6$ molecules per second per enzyme.^{4,25} In contrast, the rate-limiting step in the hydration catalytic cycle of CA mimics is the bicarbonate release step (step 4 in Scheme 1 highlighted in red). This is due to the favorable low energy of the bidentate bicarbonate-bound intermediate, and has been observed in mimics using macrocycle ligands.^{9–11,17,20,27} The consequence of the CA mimics' rate-limiting step is that they are susceptible to product inhibition by the formed bicarbonate ion, thus decreasing the hydration catalytic rates.^{9–11,17,20,27} In contrast, the surrounding amino acid environment in the natural enzyme's active site likely facilitates bicarbonate release and an easier water approach by forming an extensive hydrogen-bonding network.^{28–30}

For the application of industrial-scale CO₂ capture from the atmosphere, efficient and environment-friendly catalysts are needed.^{3,7–15} To generate the next generation of CA mimics, the intrinsic product inhibition of the hydration reaction needs to be understood before this obstacle can be overcome. Given the reversibility of the mechanism, it is not clear if the product inhibition is a consequence of an increase in the dehydration reaction, favored by the presence of the bicarbonate complex, or caused by a strong affinity of bicarbonate to the CA mimetic. To address this, more insight into the dehydration reaction kinetics is needed. With the emphasis of research focusing on CO₂ hydration catalysts,^{3,7–15} there is however a huge void of understanding concerning the dehydration of HCO₃[–] by these CA mimics and their rate-limiting step.^{18,22,31–37}

Here, we present an extensive characterization of the often-ignored dehydration reaction for the fastest small-molecule water-compatible CA mimic (ZnL1S). Kinetic measurements using stopped-flow and other relevant techniques were employed, providing the necessary insights to improve the next generation environment-friendly CO₂ capture catalysts. We report the rate-limiting step for the dehydration reaction by a CA mimic for the first time. Furthermore, this study achieves the highest reported activities by a CA mimic. Finally, we show that under specific conditions both the dehydration and the hydration reaction activities can be improved significantly beyond what is currently reported. Thus, ZnL1S has the potential to be the fastest mimic of CA known to date. Following the evidence of the bicarbonate inhibition, this work concludes with a comparison between the mimic and the enzyme, including their analysis methodologies, and consequently the true potential of ZnL1S is revealed.

EXPERIMENTAL SECTION

Experimental information including instrument details, compound characterization, and reagent sources is detailed in Supporting Information 1a.

Synthesis of the Ligand L1S and Its Metal Complexes. The ligand L1S was synthesized using an adaption of the literature method³⁸ (detailed in Supporting Information 1b), which resulted in an improved purified yield (92% yield) and reaction time (overnight) from the original synthesis procedure for the ligand and complex by Nakata et al.²⁴ In brief, 3,4-diaminobenzenesulfonic acid was reacted with nitrilotriacetic acid in a double condensation reaction in ethylene glycol to form the ligand L1S. The complexes (ZnL1S, CuL1S, NiL1S, and CoL1S) were synthesized by reacting the ligand L1S with the appropriate hydrated metal perchlorate salt in water and precipitating upon the addition of ethanol. See Supporting Information 1c for synthetic details and characterization.

Kinetic Measurement of CO₂ Hydration and HCO₃[–] Dehydration. The hydration and dehydration rate measurements were undertaken by detecting the associated pH change through the use of an appropriate colorimetric pH indicator via stopped-flow equipment (Applied Photophysics SX-20 stopped-flow spectrometer).³⁹ The pH time-dependence of CO₂ hydration or HCO₃[–] dehydration was observed via the absorbance change of a buffered indicator solution to measure the change in $V_{\text{hydr/dehydr}}$ over a concentration range of catalyst, and to determine the observed rate constant (k_{obs}) for each reaction.

Following the previously reported procedure,²⁴ a 0.1 M buffer solution containing 10×10^{-5} M indicator, 0.2 M NaClO₄, and varying concentrations of the zinc complex (ZnL1S) in water was loaded into one syringe and a CO₂ or HCO₃[−] solution in the other. CO₂ saturated solutions (33.8 mM) were prepared by bubbling CO₂ gas through water for 1 h at 25 °C. The saturated solution was diluted appropriately to give the desired concentration of CO₂. HCO₃[−] solutions were prepared by dissolving NaHCO₃ in water to give the desired concentration of HCO₃[−]. The buffer–indicator pairs were chosen on the basis of their similar pK_a's, and the pH of the buffer–indicator solution was adjusted to match these pK_a's to maximize sensitivity to pH change (Table S1).

Upon mixing of the two syringes, the change in absorbance over time was measured at 25 °C. For each buffer solution employed over the tested pH range, the buffer factor (change in absorbance vs concentration of HCl/NaOH) and linearity of CO₂ and HCO₃[−] (change in absorbance vs concentration of CO₂/HCO₃[−]) were determined as controls (Figures S5 and S6 for the buffer factor and substrate linearity, respectively). On the basis of the linearity of the response between CO₂ or HCO₃[−] concentration and change in absorbance, appropriate concentrations of CO₂ or HCO₃[−] were chosen within the linear range for the subsequent complex runs. At each concentration of complex, the V_{hydr} (rate of hydration) and V_{dehyd} (rate of dehydration) were measured, and k_{obs} (observed rate constant) was calculated as per the procedure of Nakata et al.²⁴ It is of note that not all of the observed rate constants for hydration were obtained from a linear correlation ($R^2 > 0.9$) between catalyst concentration and rate. In contrast, all dehydration k_{obs} values were obtained from linear plots. k_{cat} (catalytic rate constant) was calculated as the pH-independent catalyst rate constant as per Nakata et al. and Koziol et al. for the hydration reaction ($k_{\text{cat}} = k_{\text{obs}} \times (([\text{H}^+]/K_a) + 1)$),^{20,24} and Sun et al. for the dehydration reaction ($k_{\text{cat}} = (k_{\text{obs}} \times ([\text{H}^+] + K_a))/[\text{H}^+]$).³⁵ Each run of three repeats was undertaken three times and the average taken, to give $N = 9$.

NMR Studies of Bicarbonate-Bound Intermediate.

Studies of the catalysis mechanism of reaction were carried out and monitored via ¹H and ¹³C NMR at 25 °C. For the dehydration reaction using ¹³C-labeled sodium bicarbonate, three solutions were prepared in NMR tubes: 0.01 M complex (ZnL1S), 0.01 M substrate (NaHCO₃), and 0.01 M complex mixed with 0.01 M substrate (ZnL1S + NaHCO₃). All three solutions were made up twice, once in deuterated DMSO and once in D₂O (both H₂O-free). The solutions were then analyzed by NMR. For the hydration reaction using ¹³C-labeled CO₂, 1 atm (approximately 0.2 mmol) of ¹³CO₂ was added to the appropriate Young's tap NMR tubes using a Toepler pump. As before, the three solutions (ZnL1S complex, CO₂ substrate, and ZnL1S complex + CO₂ substrate) were made up twice, once in deuterated DMSO and once in D₂O. The subsequent hydration NMR studies were carried out in D₂O with increasing concentrations of NaHCO₃ or ¹³C-labeled NaHCO₃ (0.001, 0.01, and 0.1 M) and one concentration of NaNO₃ (0.01 M) in the appropriate Young's tap NMR tube containing 0.01 M ZnL1S and 1 atm (approximately 0.2 mmol) of ¹³CO₂. The solutions were all analyzed using a Bruker AMX-400 spectrometer at room temperature.

4-Nitrophenyl Acetate Hydrolysis Assay. The hydrolysis of 4-nitrophenyl acetate to 4-nitrophenolate with the addition of ZnL1S was monitored following the previously

reported procedure at 25 °C.⁴⁰ 50 μL of 1 mM ZnL1S (100 mM TAPS pH 8.2, 200 mM NaClO₄) was incubated with varying concentrations of CO₂ or NaHCO₃ for 5 min in a 96-well microplate. 50 μL of 5 mM 4-nitrophenyl acetate was added to start the reaction. The formation of 4-nitrophenolate was monitored at 348 nm in a Thermo Scientific Varioskan Flash Multimode microplate reader for 2000 s. Final concentrations of CO₂ or HCO₃[−] in 100 μL were 0, 1.69, 3.38, 6.76, 10.14, 13.52, and 16.9 mM. The Δabsorbance/min was calculated by subtracting the final absorbance from the starting absorbance and then dividing by the rate constant as calculated from the slope of time against absorbance.

RESULTS AND DISCUSSION

Synthesis and Catalytic Validation of ZnL1S. In this study, the previously reported fastest water-compatible small-molecule CA active site mimic ZnL1S, which uses the nitrilotris(2-benzimidazolylmethyl) ligand,²⁴ was successfully synthesized and its catalytic properties validated. The first step in the generation of the water-soluble metal complex ZnL1S was the synthesis of the ligand, L1S, as described previously in the Experimental Section and in more detail in the Supporting Information (Scheme S1 and Figures S7 and S8). The L1S ligand was then complexed with zinc as described in the Experimental Section. The synthesized ligand and the resulting complex, ZnL1S, were fully characterized by ¹H and ¹³C NMR spectroscopy, mass spectrometry, and elemental analysis (Scheme S2 and Figure S9). It is of note that, due to the new ligand synthesis procedure, as described in the methods, only the 665 isomer was seen in the product (whereas Nakata et al. saw the 665 and 666 isomers²⁴), which is of a benefit for characterization and subsequent studies. See Chart S1 for structures.

Previously, CO₂ hydration rates of ZnL1S were measured at temperatures of 5, 10, and 15 °C. These data were used to extrapolate rates to 25 °C, resulting in a predicted catalytic rate constant of 10 000 M^{−1} s^{−1}.²⁴ While this predicted rate constant makes ZnL1S the fastest reported hydration small-molecule CA mimic, there is clearly a lack of experimental data at “greener” ambient temperatures, that is, at 25 °C, which would require no heating or cooling inputs during the industrial process. Interestingly, the dehydration properties of ZnL1S have been previously studied at 25 °C at pH 6 (k_{cat} of 363 M^{−1} s^{−1}),³⁷ however, only using the less sensitive and less accurate weight loss method, and not the stopped-flow method more commonly reported for the hydration reaction.

To address this, both the CO₂ hydration and the HCO₃[−] dehydration of ZnL1S were analyzed using the stopped-flow method up to 25 °C (Table S2). We found a rate constant for CO₂ hydration of 9358 M^{−1} s^{−1} for ZnL1S at 25 °C, which is within reasonable agreement with the previously predicted value (10 000 M^{−1} s^{−1}).²⁴ For the HCO₃[−] dehydration reaction, we determined a catalytic rate constant of 103.1 M^{−1} s^{−1} at 25 °C, making ZnL1S the fastest small-molecule CA mimics for HCO₃[−] dehydration reported to date. It should be noted that the dehydration rate constant for ZnL1S had been previously measured using the less accurate weight loss method, and a catalytic rate constant of 363 M^{−1} s^{−1} was found.³⁷ This variation in measured values can be accounted for by the difference in accuracy of the methodology employed. It is of note that there was agreement with the literature for the hydration reaction as the same stopped-flow methodology was utilized. In conclusion, ZnL1S is confirmed to be the fastest

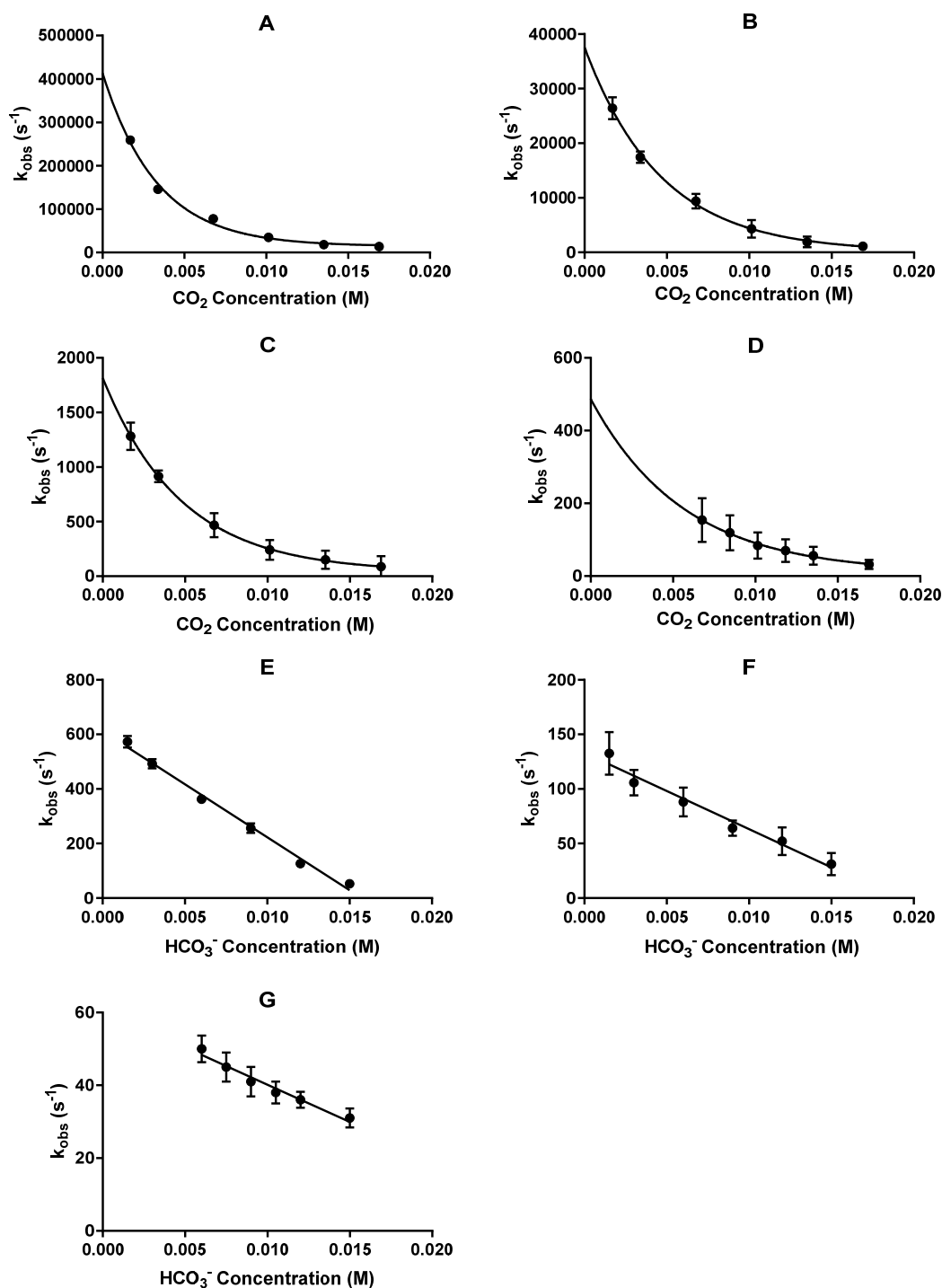


Figure 1. Nonlinear relationship between the k_{obs} of ZnL1S for CO₂ hydration across the studied pH range (A–D, pH 10.8, 9.5, 8.2, and 7.4, respectively). At pH 7.4 (D), concentrations below 0.00676 M CO₂ were below the limit of detection. The linear relationship between k_{obs} of ZnL1S (0.5, 5, and 50 μM ZnL1S) for HCO₃⁻ dehydration across the studied pH range (E–G, pH 6.3, 7.4, and 8.2, respectively). At pH 8.2 (G), concentrations below 0.005 M HCO₃⁻ were below the limit of detection. k_{obs} data are the slopes of the data shown in Figures S10B,D,F,H and S11B,D,F using 0.5, 5, and 50 μM ZnL1S (and 500 μM ZnL1S at pH 10.8). Error bars are the errors associated with the fit to generate k_{obs} .

water-soluble small-molecule CA mimic for both hydration and dehydration reactions.

Effect of Substrate Concentration and pH on ZnL1S CA Activity. To investigate the rate-limiting step of the dehydration reaction, the kinetic properties of ZnL1S for both reactions (dehydration and, for comparison, hydration) were

studied across multiple pH values and substrate concentrations.

The dehydration reaction showed a linear decrease in rate with increasing HCO₃⁻ concentration (Figure S11). Plotting the ratio $V_{\text{dehyd}}/\text{HCO}_3^-$ against catalyst concentration generated linear trends for all HCO₃⁻ concentrations.

Table 1. k_{obs} Values for the Hydration of CO_2 and the Dehydration of HCO_3^- by ZnL1S Across the Studied Range of pH Values and Substrate Concentrations^a

		pH 6.3 k_{obs} ($\text{M}^{-1} \text{s}^{-1}$)	pH 7.4 k_{obs} ($\text{M}^{-1} \text{s}^{-1}$)	pH 8.2 k_{obs} ($\text{M}^{-1} \text{s}^{-1}$)	pH 9.5 k_{obs} ($\text{M}^{-1} \text{s}^{-1}$)	pH 10.8 k_{obs} ($\text{M}^{-1} \text{s}^{-1}$)
hydration [CO_2] (M)	0.00169	below LOD	below LOD	1282 ± 125.4	26426 ± 2021	258753 ± 4242
	0.00338	below LOD	below LOD	916.1 ± 54.16	17465 ± 1044	145698 ± 2563
	0.00676	below LOD	154.10 ± 60.48^b	466.6 ± 110.1^b	9397 ± 1358	77679 ± 2969
	0.00845	below LOD	119.30 ± 48.41^b	not measured	not measured	not measured
	0.01014	below LOD	84.48 ± 36.36^b	239.8 ± 90.00^b	4333 ± 1609^b	34739 ± 3224
	0.01183	below LOD	70.26 ± 30.59^b	not measured	not measured	not measured
	0.01352	below LOD	56.05 ± 24.79^b	150.5 ± 83.00^b	1911 ± 962.5^b	18157 ± 1546
	0.0169	below LOD	31.92 ± 12.84^b	86.88 ± 97.82^b	1105 ± 769.7^b	13346 ± 1545
dehydration [HCO_3^-] (M)	0.0015	572.7 ± 21.39	132.5 ± 9.56	below LOD	below LOD	below LOD
	0.003	491.6 ± 16.89	105.7 ± 11.57	below LOD	below LOD	below LOD
	0.006	361.9 ± 15.77	88.02 ± 13.25	49.53 ± 3.689	below LOD	below LOD
	0.0075	not measured	not measured	45.36 ± 3.865	below LOD	below LOD
	0.009	255.6 ± 17.97	63.53 ± 7.114	40.77 ± 4.061	below LOD	below LOD
	0.0105	not measured	not measured	38.22 ± 3.140	below LOD	below LOD
	0.012	126.1 ± 7.998	51.90 ± 12.82	35.67 ± 2.245	below LOD	below LOD
	0.015	51.88 ± 0.7247	30.53 ± 10.26	31.14 ± 2.601	below LOD	below LOD

^aThe data are the average of three repeats. ^bIndicates that the plot of $V/[\text{CO}_2]$ versus $[\text{ZnL1S}]$ for that [substrate] had an $R^2 < 0.9$. LOD = limit of detection. k_{obs} data are the slopes of the data shown in Figures S10B,D,F,H and S11B,D,F using 0.5, 5, and 50 μM ZnL1S (and 500 μM ZnL1S at pH 10.8). Extra substrate concentrations were studied for pH 7.4 hydration and pH 8.2 dehydration to compensate for the loss of data points at lower substrate concentrations.

Additionally, there was an inhibitive effect with increasing substrate concentration, and the effect was greater at the higher concentration of catalyst (as seen by the greater slope at higher mimic concentrations in Figure S11). Furthermore, the k_{obs} decreased linearly with increasing substrate concentration (Figure 1E–G).

For the hydration reaction, there was an exponential decrease in rate with increasing CO_2 concentration, across all conditions (Figure S10). The inhibitory effect was greater at higher concentrations of catalyst (as seen by the greater slope at higher mimic concentrations in Figure S10). Additionally, the catalyst's observed rate constant (k_{obs}) decreased exponentially with increasing CO_2 concentration (Figure 1A–D). Upon plotting of $V_{\text{hyd}}/\text{CO}_2$ against catalyst concentration, the linearity of this trend improved with decreasing concentrations of CO_2 , indicating less bicarbonate inhibition (i.e., the R^2 value of the linear plot increased toward 1, Table S3). The catalysts appear more inhibited by bicarbonate with increasing catalyst concentration, due to the subsequent increased bicarbonate produced. Thus, there is a decrease in the expected improvement in activity with increased catalysts concentration, resulting in a plateau and decreasing linearity. A significant improvement in linearity was seen, however, at pH 10.8. This change in inhibition at pH 10.8 is likely the result of the easier release of carbonate, which is 3 times more abundant than bicarbonate at this higher pH. This was previously reported for a different CA mimic,¹¹ thus validating the results seen.

The differences between the effect of substrate concentration on the hydration or dehydration reactions suggest that the subsequent bicarbonate inhibition is more prevalent in the hydration reaction than in the dehydration reaction. This is evident by the exponential slope for the hydration reaction as compared to the linear slope for the dehydration reaction (Figure 1). This is the first time any inhibition for the dehydration reaction by a CA mimic has been observed.

Table 1 summarizes the calculated k_{obs} values for both reactions across the pH and substrate concentration ranges

studied. As mentioned previously, a decrease in k_{obs} is observed with increasing substrate concentration for both reactions. Additionally, the expected increase in hydration k_{obs} and expected increase in dehydration k_{obs} with increasing and decreasing pH, respectively, were seen. As generally observed in previous studies,^{18,33,34,37} the hydration reaction was faster than dehydration at pH 8.2 and above, whereas the dehydration reaction was faster at pH 6.3. Interestingly, at pH 7.4, the two reactions had very similar observed rate constants at higher substrate concentrations. The hydration reaction was, however, slightly faster at the same substrate concentration. This is the first time that a CA mimic has been shown to have similar observed rate constants for both reactions at the same pH value. Furthermore, this is a change from the enzyme. CA only exhibits faster catalysis for the dehydration reaction over hydration at pH's below 7,^{17,22,34,36,41} whereas the mimic ZnL1S demonstrates this at neutral pH.

In previous studies,^{18,22,33–37} much larger HCO_3^- concentrations were used (of the order of $\times 10^{-3}$ M), and as the data here show, this leads to catalyst inhibition. Therefore, previous studies concluded that hydration is much faster at pH 7.4. This study shows that under certain substrate conditions (i.e., at higher substrate concentrations), however, it is possible for ZnL1S to catalyze the dehydration reaction faster than the hydration reaction at neutral pH values. This conclusion could likely be applied to other CA small-molecule mimics as they share the same mechanism. The specific substrate concentrations required would vary depending upon the catalysts' affinity for bicarbonate.

On the basis of its mechanism, it is expected that the relationship between hydration k_{cat} (the catalyst's rate constant independent of pH) and H^+ concentration should be linear. This is due to the activation of the catalyst with increasing pH as the zinc-bound water molecule is deprotonated (Scheme 1, step 1).³³ Surprisingly, however, this relationship instead followed a power regression model (Figure S12A). The nonlinearity of this trend suggests that there is a loss of

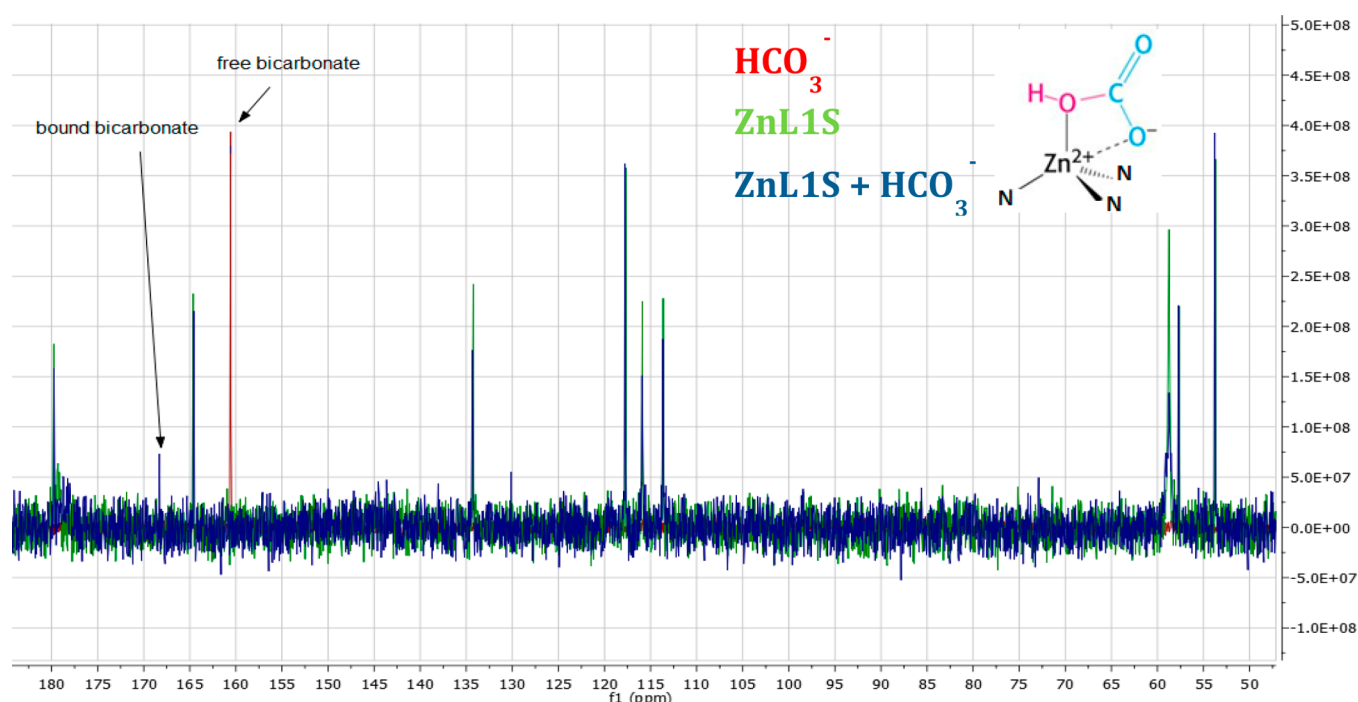


Figure 2. ^{13}C NMR spectrum in DMSO comparing free bicarbonate, ZnL1S, and ZnL1S with bicarbonate illustrating the ability of ZnL1S to bind to bicarbonate. Free bicarbonate is seen at 160 ppm and bound bicarbonate at 167 ppm. ZnL1S peaks at 180, 165, 135, 117, 116, 114, 59, and 54 ppm. Peak at 57 ppm is due to ethanol used to enhance the solubility of sodium bicarbonate with the complex in *d*-DMSO.

inhibition with decreasing H^+ concentration (increasing pH). This leads to greater improvement of the k_{cat} values than expected from activation of the catalyst alone. At higher pH values, as well as the expected increase in hydration-active OH^- -bound catalyst, there is an increased conversion to the weaker binding carbonate over bicarbonate, resulting in a faster release of product and thus decreasing inhibition.

In contrast, the catalyst's HCO_3^- dehydration activity increased linearly with increasing H^+ concentration (decreasing pH) (Figure S12B). This agrees with the catalyst's mechanism and previous investigations of other mimics, as more dehydration-active H_2O -bound catalyst was formed.^{22,35,36} We found that bicarbonate inhibition was not a significant limiting factor affecting the dehydration reaction. This contrasts with the hydration reaction, which was significantly impacted by the produced bicarbonate concentration, as seen by its nonlinear slopes in Figures 1 and S12.

The results thus far have shown that the catalyses of both the hydration and the dehydration reactions, by ZnL1S, are inhibited at higher concentrations of substrate. As pointed out previously (see Introduction), the CO_2 hydration by ZnL1S and other mimics is rate-limited by the bicarbonate-bound state of the catalysts.^{9–11,17,20,27} The rate-limiting step of HCO_3^- dehydration is however currently unknown. Given the natural equilibrium between HCO_3^- and CO_2 , it is also unclear which species is responsible for the observed inhibition of the dehydration reaction.

Capturing the Bicarbonate-Bound Intermediate via NMR. To provide evidence for the dehydration reaction's rate-limiting step, an NMR study was undertaken to see if a bicarbonate-bound intermediate could be captured. Employing the Zn^{2+} complex in aqueous and nonaqueous solvents (D_2O and/or deuterated DMSO), both the hydration and the dehydration reactions were studied. The study also allowed for a comparison of each reaction's NMR profile in each solvent.

There was no bound bicarbonate intermediate detected as part of the dehydration reaction in D_2O , due to the speed of the reaction relative to a typical NMR scan duration. It was apparent, however, that the reaction had proceeded to completion, as observed through a loss of HCO_3^- and production of CO_2 and H_2O (Figures S13 and S14). In DMSO, however, a bound Zn^{2+} bicarbonate intermediate (Figure 2), as well as the production of CO_2 and H_2O (Figures S15–17), could be observed as the mimic's turnover was slowed in this solvent. This suggests that the rate-limiting step of the dehydration reaction is also at the bicarbonate-bound intermediate stage, as is the case with the hydration reaction.

Evidence of the hydration reaction was not detected in DMSO (Figures S18 and S19), due to the inability of the bound water molecule to deprotonate as per the first step of the catalyst's mechanism (Scheme 1), and as seen for a different CA mimic.⁴² In D_2O , however, production of free HCO_3^- by the hydration reaction was observed, although the intermediate of the hydration reaction was not detected. The latter is not surprising, given the fast reaction times for the hydration reaction as compared to the speed at which a NMR scan reaches completion (Figures S20 and S21).

To visualize the bicarbonate-bound intermediate through NMR during the hydration reaction, attempts were made to slow the reaction through the addition of the hydration product HCO_3^- . By adding HCO_3^- into the hydration reaction as catalyzed by ZnL1S, the activity of the catalyst should slow as the equilibrium is pushed back toward the substrate CO_2 . This effect on the hydration rate by addition of HCO_3^- has been seen previously with a different CA mimic, although not via NMR but through the observed effect on the mimic's kinetics via stopped-flow.¹¹ Furthermore, the addition of a product mimic, such as NO_3^- , should inhibit ZnL1S sufficiently to allow the bicarbonate-bound intermediate to be captured. With increasing concentrations of HCO_3^- , evidence

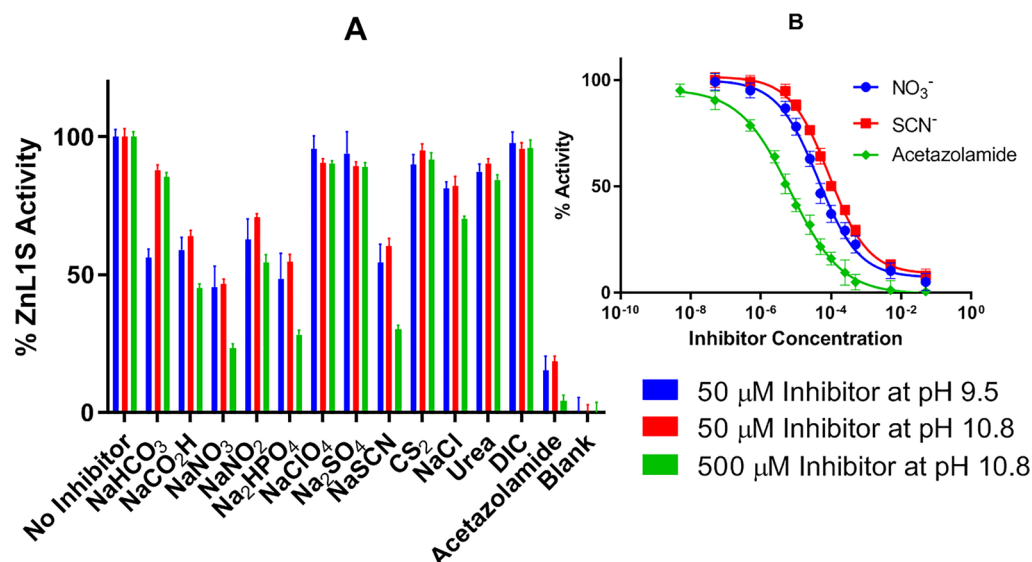


Figure 3. (A) The effect of different substrate mimics and known CA inhibitors on the % activity of CO₂ hydration by 50 μM ZnL1S over a range of pH and inhibitor concentrations. (B) The IC₅₀ curves for the inhibitors NO₃⁻, SCN⁻, and acetazolamide on the activity of CO₂ (1.69 mM) hydration by 5 μM ZnL1S at pH 10.8. All data are the average of three repeats of three. Error bars are the standard deviation (*N* = 9).

of increased zinc-bound bicarbonate was seen with an increasing upfield shift of the aromatic ¹H NMR peaks of the complex ligand (Figure S22). Additionally, upon introduction of NO₃⁻, a slight upfield shift was again seen, indicating zinc-bound bicarbonate or nitrate, although it is unknown which one is bound. Regardless, this demonstrates that when the hydration reaction is slowed, the intermediate rate-limiting bicarbonate-release step can be captured.

In summary, it was shown that the catalyst behaves as expected by undergoing both reactions under the appropriate conditions. Most importantly, however, evidence of the intermediates of both reactions was captured demonstrating the complex's ability to strongly bind bicarbonate.

Effect of CO₂ Hydration Substrate and Product Mimics on ZnL1S CO₂ Hydration Activity. To further confirm the likely inhibitory species (CO₂ or HCO₃⁻), the inhibitory effect of CO₂ and HCO₃⁻ mimics on ZnL1S hydration activity was undertaken. It is well established that CA enzymes can be inhibited by a range of anions such as Cl⁻, Br⁻, and NCS⁻. These anions bind via replacement of the coordinated water molecule or via direct insertion into the active site, thus increasing the coordination number around the zinc center.^{21–24} They have previously been tested on other CA mimics, such as the well-studied mimic zinc cyclen as well as ZnL1S.^{9,10,17,24}

In this study, a wider range of potential inhibitors were chosen, including those previously studied. Potential inhibitors were selected that are structurally and electronically related to either CO₂ or HCO₃⁻, the reactants for the CA-catalyzed reactions. The commercially available CA inhibitor acetazolamide was also tested to compare its binding to the enzyme and its synthetic mimic. Charge, geometry, and localization of electrons were all key factors in choosing the inhibitors employed (see Table S4 for details). As can be seen from the data presented in Figure 3, neutral species such as the linear CS₂ or bulky DIC (both resembling CO₂) and trigonal planar urea were not inhibitory. Charged tetrahedral species showed a mixed response. Phosphate was the only tetrahedral species revealing significant inhibition, whereas sulfate and perchlorate

did not inhibit the ZnL1S catalyzed CO₂ hydration. Single-charge inhibitors with planar geometry (e.g., chloride, thiocyanate, nitrite, bicarbonate, formate, and nitrate) were all inhibitory, although nitrate and thiocyanate were the most potent. Interestingly, the well-known sulfonamide CA enzyme inhibitor acetazolamide was the most effective inhibitor of the tested compounds.

The inhibitory constants for the most potent ionic inhibitors, nitrate and thiocyanate, were therefore determined. The IC₅₀ for NO₃⁻ of 4.236×10^{-5} M is about 2 times smaller than that observed for SCN⁻, 8.791×10^{-5} M. A two-sample *t* test (16) = 27.34, *p* = <0.0001 shows that there is a statistically significant difference between the two IC₅₀ values. Both NO₃⁻ and SCN⁻ share a single negatively charged planar geometry, which may reflect the transition state of the Zn-bicarbonate complex (see Scheme 1 between steps 3 and 4). In particular, NO₃⁻ shares charge and geometry with HCO₃⁻, and hence has been used previously to explore bicarbonate binding to CA mimics.⁴³ SCN⁻ shares only its linear shape with CO₂, and thus may well comprise the charge of the HCO₃⁻ with the shape of the CO₂, making it a transition state mimic. Because the inhibitory potency correlates with its affinity for the catalyst, one can conclude that bicarbonate mimicking inhibitors (e.g., nitrate) are better in binding to the catalyst than the carbon dioxide mimicking test compounds (e.g., CS₂; DIC). This in turn indicates a preference of the ZnL1S catalyst for binding bicarbonate over CO₂. This inhibitor profiling demonstrates that bicarbonate mimics are stronger inhibitors of, and thus bind more strongly to, ZnL1S than CO₂ mimics. This corroborates the data presented so far and provides further evidence that the bicarbonate-bound intermediate is the most likely rate-limiting step. This is in agreement with similar smaller investigations for other CA mimics.^{9–11,17,20,27} The IC₅₀ and *k_i* values for the CA enzyme inhibitor acetazolamide were determined for the inhibition of the CO₂ hydration reaction (6.99×10^{-6} and 3.14×10^{-6} M, respectively). This is evidence that the Zn-complex, without the surrounding amino acids in the enzyme's active site, is

sufficient for the interaction of this important drug molecule within the catalytic cavity of CA enzymes.^{24,44–46}

The effect of pH (9.5 versus 10.8) was examined via the inhibition of ZnL1S by the tested compounds to see how it affects the inhibitor profile. Specifically, the effect that the ratio of bicarbonate to carbonate has on either side of their pK_a of 10.3 was observed. Only NaHCO_3 showed a significant effect, with a stronger inhibitory potency at pH 9.5 than at pH 10.8. The increased concentration of the weaker binding carbonate, over the stronger binding bicarbonate, at higher pH was the reason for this difference, as discussed earlier.

The effect of inhibitors on the dehydration reaction (Figure S23) yields results similar to those of the hydration reaction. It was observed that those compounds that most similarly mimic bicarbonate (i.e., charged species) inhibit ZnL1S dehydration most significantly at pH 6.3. Interestingly, the addition of CO_2 resulted in a decrease in activity of 27%. This can be attributed to the push back of the equilibrium by introducing the product into the reaction mixture, resulting in slower dehydration activity. Overall, the dehydration reaction seems to be more susceptible to inhibition by these mimics than the hydration reaction, although the sequence of inhibitor strength and compound is the same for both reactions.

Following this, the study of the effect of CO_2 and HCO_3^- mimics on the typical CA reactions and the effect of CO_2 and HCO_3^- on non-CA catalysis by ZnL1S was explored for further evidence of the rate-limiting species.

Effect of CO_2 and HCO_3^- on ZnL1S Esterase Activity.

It has been well documented that CA synthetic mimics demonstrate esterase activity by catalyzing the hydrolysis of esters, typically 4-nitrophenyl acetate, into their respective alcohols and carboxylic acids (or their respective conjugate bases), although at much lower rates than their CO_2 hydration and HCO_3^- dehydration activities.^{23,46–48} The effect of increasing concentrations of either CO_2 or HCO_3^- on the esterase activity of ZnL1S (Figure 4) (measured by the

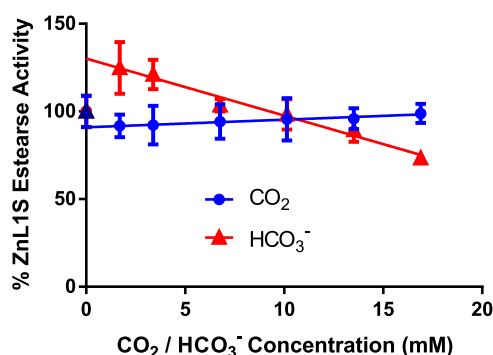


Figure 4. Effect of addition of CO_2 and HCO_3^- on the hydrolysis of 4-nitrophenyl acetate by 500 μM ZnL1S. Reactions were monitored at 348 nm. Data are the average of three repeats of three. Error bars represent the standard deviation ($N = 9$). The linear fit excludes 0 mM $\text{CO}_2/\text{HCO}_3^-$.

conversion of 4-nitrophenyl acetate into 4-nitrophenolate and acetate) provides an alternative method of finding evidence for the rate-limiting step. CO_2 showed no significant inhibition, whereas HCO_3^- inhibited the esterase activity with increasing concentrations from 1.69 to 16.9 mM HCO_3^- . There was an initial activity rise from the associated increase in pH from bicarbonate. The results demonstrate that HCO_3^- binds to the zinc center, causing an inhibition of the esterase activity of the

catalyst. In contrast, no binding was observed for CO_2 ; thus these data support the conclusion drawn from the NMR experiments described above, and provide evidence for previous predictions made by other investigations for both ZnL1S and other CA mimics.^{9–11,17,20,27} Interestingly, this is the first time bicarbonate inhibition of esterase activity has been shown for ZnL1S, which complements similar data obtained with a different CA mimic.⁴⁶

Effect of Metal Center. The experimental evidence in this study has shown that the rate-limiting step for both the hydration and the dehydration reactions by ZnL1S is at the bicarbonate-bound intermediate step. To better understand the reasons behind the mechanism, and thus to see if the preference for either reaction can be changed, we investigated the role of the metal center on the rate of hydration and dehydration of ZnL1S. To achieve this, the L1S ligand was complexed to several different transition metals, Cu, Ni, and Co, instead of Zn (Scheme S3). While previous investigations with other mimics confirmed zinc's superiority for catalytic behavior,^{16–19} the L1S ligand chelation to different metal centers had yet to be explored. Furthermore, the effect of metal center on dehydration activity has not been investigated before for any CA mimic. The three novel transition metal complexes of L1S were successfully synthesized (Supporting Information section 1c), which were subsequently tested for their ability to catalyze both the hydration reaction and the dehydration reactions.

The experimentally observed rate constants for CO_2 hydration were measured for each transition metal complex at pH 9.5, 8.2, and 7.4 using a typical CO_2 hydration measurement (see Figure S24 and Table 2 for results). As expected from the mechanism, the catalysts were most active for CO_2 hydration at pH 9.5. There was, however, a nonlinear trend between proton concentration and k_{cat} , demonstrating that all of the catalysts suffered from bicarbonate inhibition regardless of metal center. The order of activity for the transition metal L1S complexes at pH 9.5 was $\text{ZnL1S} > \text{CoL1S} > \text{CuL1S} > \text{NiL1S}$.

As predicted, ZnL1S was shown to be the fastest catalyst for CO_2 hydration, in alignment with other CA mimic studies investigating the effect of alternative metal centers.^{16–19} The higher observed rate constant for CoL1S over CuL1S and NiL1S was attributed to the similar coordination chemistries of Zn(II) and Co(II), and the fast interconversion of these two metals between four-, five-, and six-coordination states,⁴⁹ as well as the similar ionic radii (88 for Zn(II) and 88.5 for Co(II)).⁵⁰ This trend of $\text{ZnL1S} > \text{CoL1S} > \text{CuL1S} > \text{NiL1S}$ directly correlates with the likelihood of the metal centers to form bidentate coordination to bicarbonate, $\text{Ni} > \text{Cu} > \text{Co} > \text{Zn}$. As described by Han et al.,⁴³ Zn(II) derivatives bind to bicarbonate with unidentate coordination, and the Co(II) derivatives show binding that was intermediate between unidentate and bidentate. In contrast, the Cu(II) and Ni(II) derivatives show a higher tendency to exhibit bidentate coordination.⁴³ Furthermore, the similar pK_a values (8.28 for Co and Ni and 8.29 for Cu and Zn) exclude the pK_a of the metal-bound water as a determining factor for the differences seen in the interconversion of $\text{CO}_2/\text{HCO}_3^-$ between the metal complexes. The tighter binding of bicarbonate results in the slower release of product, resulting in slower rates of hydration. This is especially true at lower pH's as illustrated by the nonlinear pH dependency. Thus, the importance of over-

Table 2. Summary of k_{obs} for the Hydration of CO_2 and Dehydration of HCO_3^- by CA Mimics at Multiple pH Values^a

		metal center			
		Zn	Cu	Ni	Co
CO_2 hydration k_{obs} ($\text{M}^{-1} \text{s}^{-1}$)	pH 9.5	8803 $\pm$ 550.3	4478 $\pm$ 523.6	3489 $\pm$ 347.7	5766 $\pm$ 670.6
	pH 8.2	574.5 $\pm$ 34.59	93.6 $\pm$ 20.70	35.1 $\pm$ 15.77	102.4 $\pm$ 19.98
	pH 7.4	128.3 $\pm$ 14.43	44.3 $\pm$ 12.55	below LOD	below LOD
HCO_3^- dehydration k_{obs} ($\text{M}^{-1} \text{s}^{-1}$)	pH 6.3	102.1 $\pm$ 12.98	44.3 $\pm$ 9.91	below LOD	79.6 $\pm$ 10.08
	pH 7.4	8.5 $\pm$ 0.57	3.01 $\pm$ 0.49	below LOD	5.9 $\pm$ 0.63
	pH 8.2	5.1 $\pm$ 0.52	below LOD	below LOD	below LOD

^aThe data are the average of three repeats. k_{obs} data calculated from the fit of the data in Figures S24 and S25. LOD = limit of detection.

coming the rate-limiting step by reducing bicarbonate affinity for the metal center is emphasized.

The initial activity for HCO_3^- dehydration was measured for each transition metal complex at pH 6.3, 7.4, and 8.2 (see Figure S25 and Table 2 for results). As expected from the mechanism, the catalysts were most active for HCO_3^- dehydration at pH 6.3. The linear trend between proton concentration and k_{cat} for all active catalysts demonstrated minimal bicarbonate inhibition for these mimics. It can be observed from the data that ZnL1S is the superior catalyst for HCO_3^- dehydration at pH 6.3 and pH 7.4, followed closely by CoL1S. A similar trend in rate constant was observed at pH 6.3 and pH 7.4, as seen for the hydration reaction at pH 9.5. This also correlated with the strength of bicarbonate binding. Tighter binding would result in a slower catalytic rate, for both CO_2 hydration and HCO_3^- dehydration, and could explain why no activity was observed for NiL1S.

This study showed that the zinc metal center generated the fastest rates and k_{obs} for both the hydration and the dehydration reactions, followed by cobalt, copper, and then nickel. The zinc center was therefore key to the ability of the catalyst to efficiently hydrate CO_2 and dehydrate HCO_3^- due to its preferred weaker monodentate binding to bicarbonate. This demonstrates just how important bicarbonate affinity is in determining the mimic's activities and overcoming the rate-limiting step.

Future Improvements in Catalyst Activity through Strategic Ligand Design. As discussed, it is desired that rates as fast as that of the natural CA enzyme can be achieved by its synthetic mimics. For this goal to be reached, the effect of bicarbonate inhibition needs to be addressed. The zinc center within CA mimics is highly electron deficient and therefore easily coordinates to anions, such as bicarbonate. Although it is not the case for ZnL1S, this effect is often enhanced further in many reported CA mimic complexes, due to their cationic nature, which attract anionic species into their secondary coordination sphere to maintain charge balance, that drives the equilibrium to the bicarbonate-bound species.¹⁰ Our work has shown that the limitations of bicarbonate binding, resulting in slower activity, should be addressed in the development of future CA mimics. There are two potential ways in which this could be achieved: (i) reducing the affinity of the anion for the zinc center to discourage strong bicarbonate binding and (ii) facilitating the approach of water into the hydrophobic pocket around the zinc atom to displace the bicarbonate anion. Both can be primarily achieved through intelligent ligand design.

To reduce anion affinity for the metal center, more electron-donating ligands should be used. The ligand L1S, employed in this study, is a good example of this, as it is an anionic, water-soluble, salen-like ligand, which has the ability to donate

electron density into the metal center and therefore facilitates dissociation of bicarbonate and thus improves the rate of CO_2 hydration.^{9,10} This is part of the reason why ZnL1S is the fastest small-molecule CA mimic seen in the literature.

The other reason why ZnL1S is highly active for CO_2 hydration is that Nakata et al. designed the mimic to contain a hydrophobic pocket, which facilitates the approach of CO_2 for hydration.²⁴ The approach of water to displace the formed bicarbonate is, however, consequently unfavorable, thus rate-limiting the catalyst. To overcome this, a ligand should include the ability to be able to induce an extensive hydrogen-bonding network in the proximity of the metal center, much like the natural enzyme CA, which achieves this by utilizing three hydrogen-bonding amino acids.^{28–30} For this reason, carbonic anhydrase is limited not by bicarbonate release but by the shuttle of protons,²⁶ and therefore is significantly faster than any synthetic mimic reported to date.²⁷ With the structural changes recommended, even greater improvements in catalytic activities could be achieved.

Comparison of ZnL1S Kinetic Properties to CA and Other CA Mimics. Historically, CA mimics have struggled to match the hydration activity of the natural enzyme (e.g., hCA II hydration $k_{\text{cat}}/k_{\text{m}}$ of $1.5 \times 10^8 \text{ M}^{-1} \text{s}^{-1}$).⁵¹ Using a lower concentration of substrate (1.69 mM CO_2), we have, however, achieved a step in the right direction toward this goal. The hydration k_{cat} achieved by ZnL1S in this study is the highest reported by a small-molecule CA mimic catalyst to date. With a k_{cat} of $28\,093 \text{ M}^{-1} \text{s}^{-1}$, ZnL1S has a catalytic rate that is over 5 times faster than that of the next best small-molecule mimic, Zn Cyclam, with a k_{cat} of $5040 \text{ M}^{-1} \text{s}^{-1}$ (Table S5). By employing a lower concentration of substrate, ZnL1S was found to be almost 3 times faster than had been previously predicted from the low temperature study reported by Nakata et al.,²⁴ and thus a better idea of the true potential of the catalyst for CO_2 hydration was seen. Furthermore, it is possible that decreasing the CO_2 concentration further could improve the k_{cat} beyond the value reported here, but this concentration falls below our current methodology's limit of detection. At pH 10.8, the k_{cat} of ZnL1S is similar to that of the recent synthetic metalloenzyme developed by Zastrow et al.,⁵² which at the time demonstrated the highest recorded rate of CO_2 hydration for a CA mimic.⁵² At pH 9.5, however, the metalloenzyme was faster. This metalloenzyme is, however, significantly larger and protein-like, and therefore suffers from a lower thermal stability and more complex synthesis than a small-molecule mimic like ZnL1S.

CA mimics have also struggled to match the dehydration activity of the natural enzyme (e.g., hCA II dehydration $k_{\text{cat}}/k_{\text{m}}$ of $1.2 \times 10^7 \text{ M}^{-1} \text{s}^{-1}$).⁵³ The dehydration k_{cat} for ZnL1S described here has shown a 60% improvement of that previously reported for the mimic, as measured via the

Table 3. Comparison of Inhibition Data for the Complex ZnL1S with Data from Different Isoforms of Human CA (hCA)^a

	IC ₅₀ (μM)		K _i		
	ZnL1S		hCA I	hCA II	hCA VI
acetazolamide	6.99 ± 0.50	3.14	250 nM	12 nM	11 nM
NO ₃ [−]	42.36 ± 2.93	27.9	7 mM	35 mM	0.76 mM
SCN [−]	87.91 ± 4.05	59.8	0.2 mM	1.6 mM	0.89 mM

^aZnL1S CO₂ hydration data were measured using 5 μM ZnL1S and 1.69 mM CO₂ at pH 10.8 and 25 °C. Data are the average of three repeats. Enzyme data are from Bertucci et al. and Supuran et al., measured using 235 nM hCA I and 1.5 nM hCA II and VI, at pH 7.5 and 20 °C.^{54,55}

weight-loss method.³⁷ ZnL1S has a dehydration k_{cat} of 579 M^{−1} s^{−1}, which is over 10 times that of the next fastest mimic Zn Cyclen, with a k_{cat} of 55 M^{−1} s^{−1} (Table S5). ZnL1S has therefore shown the fastest dehydration k_{cat} reported for a small-molecule CA mimic to date. Consequently, this work represents a significant step toward matching the activity demonstrated by CA. Comparing the k_{cat} values for the hydration and dehydration reactions, ZnL1S shows preference for the hydration reaction rather than dehydration, as is the case for both the enzyme and other CA mimics published previously.

Simply by reducing the concentration of substrate used for both the hydration and the dehydration reactions, we have reported increases in catalytic rates due to weaker bicarbonate binding. This has been without the need for any modifications to the complex's structure. It is expected that this principle could be applied to any CA mimic used for CO₂ capture, which follows the same mechanism as ZnL1S and therefore has the same rate-limiting step.

Comparison of Inhibitor Effects on ZnL1S and CA. In the previously discussed inhibitor study, it was observed that the negatively charged species were more inhibitory for the mimic (micromolar range K_i) than for the enzyme (millimolar range K_i) (Table 3). It is likely that it is easier for the anions to enter the active site of the complex and compete with substrates in the more solvent-exposed center. The opposite was true for acetazolamide where it was more inhibitory for the enzyme (nanomolar range K_i) over the mimic (micromolar range K_i) (Table 3). This is probably caused by lack of acetazolamide binding amino acids for the mimic, as well as an increased steric hindrance when acetazolamide binds to ZnL1S.²⁴ The application of inhibitor profiling (which is commonly used to characterize enzymes) to probe the hydration mechanism of CA mimics revealed new insights into how binders interact differently between native CA and its mimics.

ZnL1S only showed inhibition by bicarbonate and no inhibition by carbonate, whereas it has been shown that hCA II is inhibited by both bicarbonate and carbonate, although the inhibition is significantly weaker for the enzyme (K_i = 85 and 73 mM for bicarbonate and carbonate, respectively)⁵⁶ than for ZnL1S (nearly 50% reduction in activity with 50 μM HCO₃[−]). It should be noted that there is no observable difference between inhibition of CA by bicarbonate or carbonate, as bicarbonate release is not the rate-limiting step for the natural enzyme, highlighting further the different rate-limiting steps between the native enzyme and its mimics. It is also worth noting that our data reveal a much less potent effect of chloride on ZnL1S activity than previously reported,²⁴ which could be due to the ZnL1S complex being generated in situ in the stopped-flow by Nakata et al., potentially resulting in inhibition of the initial formation of the catalyst.

Comparison of Small-Molecule Mimic and Enzyme Kinetic Analysis Methodologies. Interestingly, there is a difference observed in the literature between the analysis methods employed to study rates of catalysis by enzymes and that used for their mimics. For more information on the different analysis methodologies, see Supporting Information section 2g. In brief, the kinetic data for ZnL1S were analyzed via the typical enzyme analysis methodology generating “enzyme-like” kinetic parameters. The result of this is described below and in Figure S26 and Table S6. To avoid confusion, the kinetic parameters generated by the enzyme and catalysis analysis methods will be labeled with a superscript e (for enzyme) or c (for catalyst), respectively.

Typically, the $k_{\text{cat}}/k_{\text{m}}^{\text{e}}$ values of enzymes are compared to the $k_{\text{cat}}^{\text{c}}$ of synthetic mimics, as they have the same units (M^{−1} s^{−1}) and both quantify catalytic efficiency. Upon comparison of the $k_{\text{cat}}/k_{\text{m}}^{\text{e}}$ of ZnL1S to the $k_{\text{cat}}^{\text{c}}$ of ZnL1S, a difference in the values can be seen. For both reactions and across all conditions studied, the calculated $k_{\text{cat}}/k_{\text{m}}^{\text{e}}$ values were greater than their respective mimic-like $k_{\text{cat}}^{\text{c}}$, often by orders of magnitude. As the $k_{\text{cat}}/k_{\text{m}}^{\text{e}}$ considers the effect of substrate concentration and the subsequent inhibition, it can be said that these values demonstrate a better representation of the mimic's ability to catalyze each reaction. These values show the mimic's maximal catalytic efficiency when substrate concentration is negligible (at a particular concentration of catalyst). This is rather than the catalytic efficiency per molar of catalyst at a specific substrate concentration, as often reported.

Consequently, a comparison of the $k_{\text{cat}}/k_{\text{m}}^{\text{e}}$ of ZnL1S to that of hCA II allows for a better indication of the closeness/difference between the activities of each species. The hydration catalytic efficiency ($k_{\text{cat}}/k_{\text{m}}^{\text{e}}$) of hCA II at pH 9.0 is 1.5×10^8 M^{−1} s^{−1}, whereas the highest catalytic efficiency ($k_{\text{cat}}/k_{\text{m}}^{\text{e}}$) of ZnL1S, at a similar pH, in this study was found to be 5.9×10^5 at pH 9.5. At pH 10.8, an even greater catalytic efficiency was found (2.0×10^7 M^{−1} s^{−1}). The maximum efficiency of ZnL1S hydration, that is, with no inhibition, is therefore only 2.5 magnitude smaller than that of hCA II at a similar pH. This is a significant improvement when compared to the $k_{\text{cat}}^{\text{c}}$ calculated previously using the small-molecule methodology, which gave a value of 2.8×10^4 M^{−1} s^{−1} at pH 9.5 using 1.69 mM CO₂.

The previously discussed metalloenzyme mimic developed by Zastrow et al. had a catalytic efficiency ($k_{\text{cat}}/k_{\text{m}}^{\text{e}}$) of 1.8×10^5 M^{−1} s^{−1} at pH 9.5.⁵² ZnL1S is now over 3 times more efficient at the same pH. When not inhibited by bicarbonate, ZnL1S therefore has the potential to be the fastest mimic of CA ever synthesized. Furthermore, the $k_{\text{cat}}/k_{\text{m}}^{\text{e}}$ of ZnL1S increased with decreasing concentration of mimic, and therefore the maximal catalytic efficiency could be even greater than reported here. A lower concentration of mimic could not be measured in this study due to the limit of detection of our current experimental methodology. Nevertheless, it is entirely

feasible that with a lower concentration of mimic, catalytic efficiencies closer to that of CA could be achieved.

The same conclusions can be drawn for the dehydration reaction. The dehydration catalytic efficiency ($k_{\text{cat}}/k_{\text{m}}^{\text{e}}$) of hCA II is $1.2 \times 10^7 \text{ M}^{-1} \text{ s}^{-1}$ at pH 5.5,⁵³ whereas the highest catalytic efficiency ($k_{\text{cat}}/k_{\text{m}}^{\text{e}}$) of ZnL1S in this study was found to be $1.7 \times 10^4 \text{ M}^{-1} \text{ s}^{-1}$ at pH 6.3. The maximum efficiency of ZnL1S dehydration, that is, with no inhibition, is therefore only 3 orders of magnitude smaller than that of hCA II. This is significantly higher when compared to the value of $k_{\text{cat}}^{\text{c}}$ calculated previously, $579 \text{ M}^{-1} \text{ s}^{-1}$ at pH 6.3 using 1.5 mM HCO_3^- . Our work has revealed that ZnL1S is the fastest small-molecule water-soluble CA mimic for dehydration reported to date. As with the case for the hydration reaction, ZnL1S has the potential to be an even greater catalyst for HCO_3^- dehydration with a lower concentration of mimic and potentially reach catalytic efficiencies similar to that of CA.

CONCLUSIONS

We report for the first time the rate-limiting step of the dehydration reaction by a CA mimic to be at the bicarbonate-bound step, much like the hydration reaction. This confirms that it is the mimic's affinity for bicarbonate that results in the previously observed hydration inhibition, and not the occurrence of the reverse dehydration reaction. Each reaction is, however, inhibited by bicarbonate by differing degrees, with the hydration reaction demonstrating significantly greater bicarbonate inhibition than the dehydration reaction. With this knowledge, the catalyst's preference for either reaction could be controlled via substrate concentration.

Consequently, a significant increase in the activity of the previously published CA mimic ZnL1S, which is the fastest small-molecule mimic published to date, was achieved without the need for modification of the catalyst structure. Rather, by simply lowering the concentration of the substrate used for either the hydration of CO_2 or the dehydration of HCO_3^- , we were able to achieve an increase in k_{cat} by a factor of 3 and 1.6, respectively. By lowering the concentration of substrate, especially for the hydration reaction, the energetically unfavorable rate-limiting bicarbonate-intermediate step was overcome due to the subsequently lower concentration of bicarbonate. This information will be critical for optimizing the use of such mimics in industrial carbon capture.

Conversely, by increasing the substrate concentration used, similar k_{obs} values could be seen for both reactions at the same pH (pH 7.4) for the first time. This opens the potential to manipulate mimics to enable them to show preference to catalyze HCO_3^- dehydration over CO_2 hydration at physiological pH values. As demonstrated by the maximal catalytic efficiencies ($k_{\text{cat}}/k_{\text{m}}^{\text{e}}$) of ZnL1S at negligible substrate concentration, the mimic could reach activities of only 2.5–3 orders of magnitude slower than that of the natural CA enzyme itself. This finding could likely be applied to any CA, and thus these mimics have the potential to demonstrate significantly higher catalytic activities in the future. With future careful ligand design, to facilitate bicarbonate release at a faster rate, even greater activities could also be reached.

Additionally, within this study, the observed rate constants ($k_{\text{obs}}^{\text{c}}$) for CO_2 hydration and HCO_3^- dehydration catalyzed by ZnL1S, CuL1S, NiL1S, and CoL1S (except ZnL1S for CO_2 hydration) were reported for the first time via stopped-flow. A comparison of the kinetic activity of this suite of CA mimics with differing metal centers revealed that ZnL1S was a superior

catalyst for both the CO_2 hydration and the HCO_3^- dehydration reactions as expected. The kinetic profile of ZnL1S for both hydration and dehydration activities was studied across a large pH, complex, and substrate concentration range at 25 °C. Consequently, substrate inhibition kinetics, caused by increased bicarbonate, were seen for the first time for a CA mimic. A comparison between analytical methodologies of catalysts and enzymes was carried out, and the same trends could be seen, and subsequent conclusions were drawn.

Finally, this work contains the largest inhibitor study of any CA mimic reported in the literature. The first IC_{50} and K_i values for the hydration of CO_2 by CA mimic inhibitors are also reported, for both anionic inhibitors and the commercial CA inhibitor acetazolamide. The use of substrate and product mimics to probe the mechanism of enzyme mimics, as shown here for CA and its small-molecule mimic ZnL1S, allows for a subsequent deeper understanding revealing differences between the substrate binding in the enzyme and its mimics, as well as the difference in mechanisms that this alludes to. In the case of the natural CA enzyme and the ZnL1S mimic, small anions were found to be more inhibiting for the mimic, suggesting a greater importance for anion binding affinity in the mechanism of CA mimics than the enzyme. This approach could be used to investigate any other chemical mimic of a biological enzyme, especially those that are important in medicine, such as is the case with the CA studied here.

Future work will look at how these findings, especially the effect of substrate concentration to manipulate preference for either reaction at physiological pH and the potential to improve catalytic activity further, can be applied to uses of CA and its mimics for wider-world applications.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acscatal.8b04077.

All ligand and complex spectra, as well as data from the stopped-flow and all other assays (PDF)

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Notes

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